

SEO Intelligence & AI Visibility Platform

Product Requirements & System Blueprint (Version 1.0)

Product Vision

Build an enterprise-grade SEO Intelligence Platform that combines:

- Google Analytics 4 (GA4)
- Google Search Console (GSC)
- AI Traffic Analytics
- Technical SEO Monitoring
- Competitor Intelligence
- GEO (Generative Engine Optimization)
- AI Visibility Tracking
- Automated Reporting
- AI-Powered Analysis
- AI-Powered QA Testing

The platform should automatically collect, analyze, audit, validate, and report SEO performance without manual effort.

Business Goals

Primary Goals

1. Eliminate manual SEO reporting.
 2. Provide real-time SEO visibility.
 3. Track AI-generated traffic.
 4. Detect SEO issues automatically.
 5. Generate actionable recommendations.
 6. Improve client reporting efficiency.
 7. Create a scalable multi-client platform.
 8. Build a future-ready GEO and AI Visibility system.
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User Journey

Step 1

User logs in using Google OAuth.

Step 2

System fetches:

- GA4 Accounts
- GA4 Properties
- GSC Properties

Step 3

User selects Project.

Step 4

Dashboard loads.

Step 5

Platform automatically:

- Syncs Data
- Generates Insights
- Detects Issues
- Creates Reports
- Runs QA Validation

Core Modules

Module 1: Authentication

Features:

- Google OAuth
- Secure Sessions
- Refresh Token Rotation
- Token Encryption
- Login Audit Logs
- Device Tracking
- Session Monitoring

Security:

- HTTP-only Cookies
- JWT
- CSRF Protection

- Rate Limiting
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Module 2: Project Management

Features:

- Multi Client Support
- Multi Property Support
- Domain Selector
- Property Selector

Example:

Agency |—— Client A | |—— Domain A | |—— Domain B | |—— Client B |—— Domain C

Module 3: Executive Dashboard

KPIs:

- Organic Clicks
- Organic Sessions
- Impressions
- CTR
- Average Position
- Conversions
- Revenue
- AI Sessions

Filters:

- Last 7 Days
 - Last 30 Days
 - Last 90 Days
 - Quarterly
 - Yearly
 - Custom Range
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Module 4: Search Console Dashboard

Metrics:

- Clicks
- Impressions

- CTR
- Average Position

Keyword Intelligence:

- Top Keywords
- Rising Keywords
- Dropping Keywords
- Lost Keywords
- New Keywords

Opportunity Engine:

- Position 4-10
 - High Impression Keywords
 - Low CTR Keywords
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Module 5: Analytics Dashboard

Metrics:

- Users
- Sessions
- Engaged Sessions
- Engagement Rate
- Avg Engagement Time
- Conversions

Channels:

- Organic
 - Direct
 - Referral
 - Social
 - Paid
 - AI Traffic
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Module 6: AI Traffic Dashboard

Platforms:

- ChatGPT
- Gemini
- Perplexity
- Claude

- Copilot

Metrics:

- AI Sessions
- AI Users
- AI Conversions
- AI Conversion Rate

Landing Pages:

- AI Traffic by Page
- AI Conversions by Page

AI Traffic Drop Detection

AI Growth Detection

Module 7: Landing Page Intelligence

Columns:

- URL
- Clicks
- Sessions
- Conversions
- Revenue
- AI Sessions
- Position

Filters:

- Blog
 - Service Pages
 - Product Pages
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Module 8: Historical Performance

Views:

- Weekly
- Monthly
- Quarterly
- Yearly
- All Time

Metrics:

- Traffic Growth
 - Ranking Growth
 - Conversion Growth
 - AI Growth
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Module 9: Technical SEO Dashboard

Core Web Vitals

- LCP
- CLS
- INP

SEO Monitoring

- Index Coverage
 - Sitemap Issues
 - Robots Issues
 - Canonical Issues
 - Redirect Issues
 - Broken Links
 - Crawl Errors
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Module 10: Schema Monitoring

Track:

- FAQ Schema
- Article Schema
- Product Schema
- Organization Schema
- Breadcrumb Schema

Validation:

- Missing Schema
 - Invalid Schema
 - Rich Result Eligibility
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Module 11: GEO & AI Visibility Dashboard

Track:

- AI Mentions
- AI Citations
- AI Overview Visibility
- GEO Score
- AI Visibility Score

Platforms:

- ChatGPT
 - Gemini
 - Perplexity
 - Claude
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Module 12: Competitor Intelligence

Track:

- Keyword Gap
 - Content Gap
 - SERP Ownership
 - Visibility Share
 - Traffic Share
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Module 13: Conversion & ROI Dashboard

Metrics:

- Leads
- Revenue
- Cost
- SEO ROI

Attribution:

Keyword → Landing Page → Lead → Sale

Module 14: Report Center

Reports:

- Weekly
- Monthly
- Quarterly
- Annual

Exports:

- PDF
- CSV
- Excel

Automatic Scheduling

Module 15: Alert Center

Alerts:

- Traffic Drops
- Ranking Drops
- Conversion Drops
- AI Traffic Drops
- Indexing Issues
- Technical SEO Issues

Delivery:

- Dashboard
 - Email
 - Slack
 - WhatsApp
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AI Agent Ecosystem

AI QA Agent

Responsibilities:

- API Testing
- UI Testing
- Mobile Testing

- Export Testing
- Authentication Testing
- Security Validation

Output:

System Health Score

AI SEO Analyst Agent

Responsibilities:

- Ranking Analysis
 - Traffic Analysis
 - Opportunity Detection
 - Recommendation Generation
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AI Technical SEO Agent

Responsibilities:

- Technical Audits
 - Index Audits
 - Core Web Vitals Audits
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AI Data Analyst Agent

Responsibilities:

- Data Validation
 - Data Accuracy Verification
 - Anomaly Detection
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AI UX Reviewer Agent

Responsibilities:

- UX Analysis
 - Mobile Review
 - Navigation Review
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AI Product Manager Agent

Responsibilities:

- Feature Review
 - Workflow Review
 - Product Quality Assessment
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AI Command Center

Natural Language Search

Examples:

Why did traffic drop?

Which pages receive AI traffic?

Which keywords are easiest to move into Top 3?

Show pages losing traffic.

Show pages with declining CTR.

Backend Architecture

Frontend

- Next.js
- TypeScript
- Tailwind
- shadcn/ui

Backend

- NestJS

Database

- PostgreSQL

Cache

- Redis

Queue

- BullMQ

Hosting

- Vercel
- Google Cloud

Monitoring

- Sentry
 - OpenTelemetry
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Data Pipeline

Google OAuth ↓ GA4 API ↓ GSC API ↓ Data Sync Engine ↓ Data Warehouse ↓ Analytics Engine ↓ Insight Engine ↓ Dashboard

Database Tables

Core

- users
- clients
- projects
- properties

Analytics

- ga4_data
- gsc_data
- keywords
- landing_pages
- conversions

AI

- ai_traffic
- ai_mentions

- ai_citations

System

- alerts
- reports
- audit_logs
- sync_logs
- insights

Automation Jobs

2:00 AM

GA4 Sync

2:30 AM

GSC Sync

3:00 AM

AI Traffic Processing

3:30 AM

Insight Generation

4:00 AM

Alert Generation

4:30 AM

QA Agent Validation

5:00 AM

Report Generation

Monitoring Center

Track:

- API Health
 - Database Health
 - Queue Health
 - Sync Health
 - Security Health
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Health Scores

- SEO Health Score
 - AI Visibility Score
 - GEO Score
 - Tracking Health Score
 - System Health Score
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Performance Requirements

Dashboard Load Time < 2 Seconds

API Response Time < 500 ms

Sync Success Rate > 99%

Data Accuracy > 99%

Mobile Responsive Across All Devices

Development Roadmap

Phase 1

- Authentication
- GA4
- GSC
- Executive Dashboard
- Reporting

Phase 2

- Technical SEO
- Alerts
- AI Traffic
- Insights

Phase 3

- Competitor Intelligence
- GEO Dashboard
- AI Visibility

Phase 4

- AI Agents
- AI Command Center
- Advanced Automation

Phase 5

- CRM Integrations
- Revenue Attribution
- Enterprise Expansion

Success Criteria

The platform should:

- Reduce reporting effort by 90%
- Provide accurate SEO data
- Detect issues automatically
- Generate recommendations automatically
- Monitor AI visibility
- Scale across multiple clients
- Remain fully automated
- Be mobile responsive
- Be enterprise ready